How the BSPD Works

The BSPD is the circuit that shuts the car off if the driver is braking hard while the motor is still putting out a lot of power. The rules say this has to happen if both are true for more than half a second, and it also has to shut off if one of the sensors gets unplugged or shorted. It can't use a microcontroller, so everything in my design is done with op-amps, resistors, and capacitors, and the op-amps are mostly being used as comparators that just say whether one voltage is higher than another.

The two sensors are a brake pressure sensor and a current sensor. The brake sensor puts out 0.5 V with no pressure and 4.5 V at 2000 psi, and it runs off the same 5 V as the rest of the board. This means if the 5 V goes up or down a little, the sensor output and my reference move together, so the pressure where it trips stays the same. I compare the brake signal to a voltage I set with a trim pot. Right now that is around 1.37 V, which is 30 bar, but that number is still a placeholder until we figure out what hard braking means for our car.

The current sensor was harder because its output sits at 2.5 V with no current and only changes 12.5 mV for every amp. At the 5 kW limit our pack is only pulling about 46 A, which is a change of less than 0.6 V. So I used a difference amplifier that subtracts the sensor's own 2.5 V reference from its output and multiplies what's left by about 5. That gives me a signal that starts at 0 V and goes up about 62 mV per amp, and I compare it to a fixed divider.

Something I didn't realize at first is that the current sensor's 2.5 V reference is made inside the sensor and doesn't follow our 5 V supply, but my threshold divider does. The 7805 can be anywhere from 4.8 to 5.2 V, which would let the trip point drift above 5 kW. So I lowered the threshold to 42 A, which is about 4.5 kW, so it stays under the limit even at the worst case. I also put an op-amp buffer on the reference wire, because before that, if just that one wire broke, the circuit would keep running and only trip at around 77 A without telling anyone.

Once I have a brake high signal and a power high signal, I need to know when both are true. I did this by averaging them with two equal resistors, so the middle point is 0 V if neither is on, 2.5 V if one is on, and 5 V if both are on. Then another comparator checks if that average is above 3.75 V, which only happens when both are on.

The half second delay is a resistor and capacitor. When both conditions are true the capacitor charges up, and when it gets to half of 5 V the last comparator trips. With 470k and 1 uF this takes about 0.33 s. I first had it at 0.47 s, but with the tolerance on the capacitor it could have gone past 0.5 s, so I made it shorter to leave room. When it trips, the output feeds back into the capacitor through a diode, so the capacitor stays charged and the circuit stays latched even after the driver lets go of the brake. The only way to reset it is turning the power off and back on.

For the fault detection, each sensor has a pull-down resistor, so if it gets unplugged the signal falls to 0 V instead of floating somewhere random. Then there are comparators checking whether each signal is below 0.26 V or above 4.74 V, which is outside anything the sensors

should ever output normally. If any of them sees a fault, it charges the same timer capacitor through a diode and a 100k resistor, so a real fault latches in about 0.09 s. The 100k is there because when the board first turns on the filter capacitors start at 0 V and look like a fault for less than a millisecond, and without it the BSPD would latch every time we turned the car on.

The output is two MOSFETs and a relay. I set it up so the relay is powered when everything is normal, which keeps the shutdown circuit closed. When it trips, the first MOSFET turns on, which turns the second one off, and the relay opens. I did it this way because if the board loses power for any reason the relay also opens, so a failure of the BSPD itself ends up shutting the car down instead of leaving it running. The relay contacts are also separate from the rest of the circuit, so nothing on the board can push power into the shutdown circuit.

The last part is the test loop, which is just a wire that goes through the current sensor a few times and connects to two banana jacks. The inspectors can push current through it to make the sensor think the car is drawing power without the car moving. Each turn multiplies the effect, so with 10 turns they would only need about 4.2 A to reach the 42 A trip point. The current sensor also has to be mounted with its arrow facing the direction the pack discharges, because if it is backwards the circuit would never see power at all and the test loop would still pass.